

Review Lecture (post Lecture 6)

September 14, 2023

Sample

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y_i	1	1	0	1	...	0	.66

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Remember that the mean of a binary variable is a proportion.

Sample versus Population

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After election day, we found that in fact 4, 999, 960 people, or roughly 70% of registered voters, turned out to vote.

Table: Population

j	1	2	3	4	...	...	...	...	7,233,584	$\bar{y}_{7,233,584}$
y_j	0	1	0	1	...	...	...	...	1	.70

Sampling Distribution of $\bar{Y}_{625}$

i	J_1	Y_1	J_2	Y_2	$\dots$	J_{625}	Y_{625}	$\bar{Y}_{625}$
	9562350	1	8763351	1	$\dots$	1294801	0	.68

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i	1		2		...	625		
s	J_1	Y_1	J_2	Y_2	...	J_{625}	Y_{625}	$\bar{Y}_{625}$
1	9562350	1	8763351	1	...	1294801	0	.68
2	5331704	0	4533839	1	...	3342359	1	.70

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2	5331704	0	4533839	1	...	3342359	1	.70
3	5129936	0	10981600	0	...	4096184	1	.75
4	803605	0	7036389	1	...	803605	0	.73
5	148567	0	3833847	1	...	4769869	1	.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	4163458	0	8384613	1	...	377981	1	.74
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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- ▶ What is the shape of the sampling distribution you drew?

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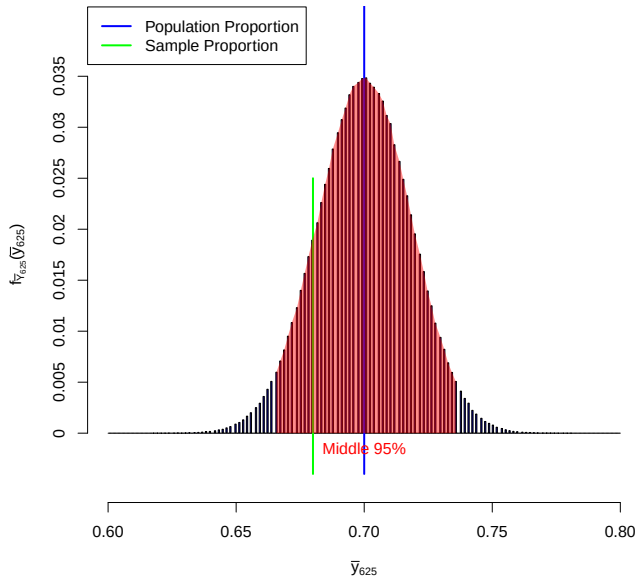
# Divide all of these numbers
# by the sample size.

Ybar_vec <- sumY_vec/625

# Plot a histogram

hist(Ybar_vec)
```

Sampling Distribution of $\bar{Y}_{625}$



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- ▶ How would you construct a 95% CI from this distribution?
- ▶ What does your answer to the larger/smaller question tell us about confidence interval coverage?
- ▶ If I tell you that the estimated standard error is 0.1, construct a 95% CI using the normal approximation formula.

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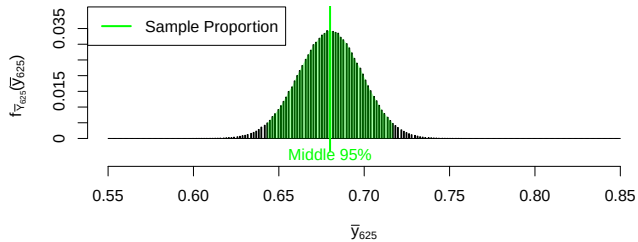
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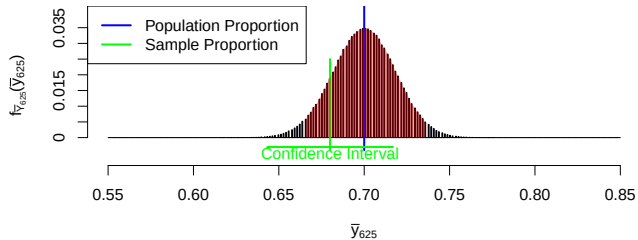
This is equivalent to replacing .70 with .68 in the R code.

```
sumY_vec <- rbinom(1000000, size=625, prob=.68)
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Estimated Sampling Distribution of $\bar{Y}_{625}$



Sampling Distribution of $\bar{Y}_{625}$



The Logic of Confidence Intervals

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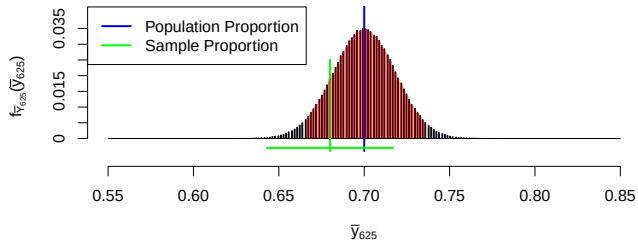
1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
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The Logic of Confidence Intervals

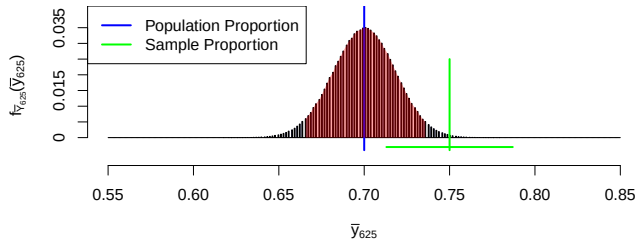
If the bootstrapped sampling distribution has the same shape as the sampling distribution,

1. The 95% confidence interval will *cover* the population proportion when the sample proportion is in the middle 95% of the sampling distribution.
2. The 95% confidence interval will *not* cover the population proportion when the sample proportion is not in the middle 95% of the sampling distribution.
3. There is a 95% probability of getting a sample proportion in the middle 95% of the sampling distribution, therefore there is 95% probability of getting a 95% confidence interval that covers the population proportion.

Sampling Distribution of $\bar{Y}_{625}$



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The Sampling Distribution of the Confidence Interval

i	1	2	...	625		
s	Y_1	Y_2	...	Y_{625}	$\bar{Y}_{625}$	95% CI
1	1	1	...	0	.68	
2	0	0	...	1	.75	
3	0	1	...	1	.70	
4	0	1	...	1	.73	
5	0	1	...	1	.69	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	
1 mil	0	1	...	1	.74	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Table: Sampling Distributions for $\bar{Y}_{625}$ and the 95% Confidence Interval

95% of the rows in this sampling distribution will have a CI that covers .70.

The Sampling Distribution of the Confidence Interval

i	1	2	...	625	$\bar{Y}_{625}$	95% CI
s	Y_1	Y_2	...	Y_{625}		
1	1	1	...	0	.68	[.643,.717]
2	0	0	...	1	.75	
3	0	1	...	1	.70	
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2	0	0	...	1	.75	[.715,.785]
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3	0	1	...	1	.70	[.663,.737]
4	0	1	...	1	.73	[.694,.766]
5	0	1	...	1	.69	[.653,.727]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	0	1	...	1	.74	[.705,.775]
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

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5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.

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2. We can observe that the bootstrapped sampling distribution of the proportion is approximately normally distributed.
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4. The variance of the sampling distribution of the proportion is $\sqrt{.7(1 - .7)/625} = .0183$.
5. The variance of the bootstrapped sampling distribution of the proportion is $\sqrt{.68(1 - .68)/625} = .0187$.
6. When n is large, these variances are close.